

## 4.1 Limits

In many situations, the value of a function at a point is less important than its behavior near that point. For example, consider  $f(x) = \frac{x^2-1}{x-1}$ ,  $x \neq 1$ . Although  $f(1)$  is not defined, we have  $f(x) = x + 1$ ,  $x \neq 1$ , so the values of  $f(x)$  become arbitrarily close to 2 as  $x$  approaches 1. Thus, the function exhibits a well-defined behavior near  $x = 1$  despite being undefined there.

More generally, when studying continuous and discontinuous functions, we are interested in understanding the behavior of a function as the input approaches a given point from the left or the right. This motivates the concept of the *limit of a function*, which provides a precise mathematical description of the value that a function approaches as its argument approaches a specified point.

**Definition 4.1.** Let  $a \in J \subseteq \mathbb{R}$  be an interval and  $f : J \setminus \{a\} \rightarrow \mathbb{R}$  be a function. We say that

$$\lim_{x \rightarrow a} f(x) = \ell$$

if for every  $\epsilon > 0$ , there exists  $\delta > 0$  such that

$$x \in J, \quad 0 < |x - a| < \delta \Rightarrow |f(x) - \ell| < \epsilon.$$

In this case, we say that  $\ell$  is the limit of  $f$  at  $a$ .

**Example 4.1.** Show that

1.  $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = 2$ .
2.  $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$ .

**Solution.**

1. Let  $\epsilon > 0$  be given. Choose  $\delta = \epsilon$ . If  $0 < |x - 1| < \delta$ , then

$$\left| \frac{x^2-1}{x-1} - 2 \right| = |x+1-2| = |x-1| < \delta = \epsilon.$$

$$\text{Hence, } \lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = 2.$$

2. Let  $\epsilon > 0$  be given and choose  $\delta = \epsilon$ . If  $0 < |x| < \delta$ , then

$$\left| x \sin\left(\frac{1}{x}\right) \right| = |x| \left| \sin\left(\frac{1}{x}\right) \right| \leq |x| < \delta = \epsilon.$$

$$\text{Hence, } \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0.$$

**Theorem 4.2.** The limit of a function, if it exists, is unique.

*Proof.* Suppose  $\lim_{x \rightarrow a} f(x) = \ell_1$  and  $\lim_{x \rightarrow a} f(x) = \ell_2$ . Assume, to the contrary, that  $\ell_1 \neq \ell_2$ . Choose  $\epsilon = \frac{|\ell_1 - \ell_2|}{3} > 0$ . Then there exist  $\delta_1, \delta_2 > 0$  such that

$$0 < |x - a| < \delta_j \implies |f(x) - \ell_j| < \epsilon, \quad j = 1, 2.$$

Let  $\delta = \min\{\delta_1, \delta_2\}$ . If  $0 < |x - a| < \delta$ , then

$$|f(x) - \ell_1| < \epsilon \quad \text{and} \quad |f(x) - \ell_2| < \epsilon.$$

Hence,

$$|\ell_1 - \ell_2| \leq |\ell_1 - f(x)| + |f(x) - \ell_2| < 2\epsilon = \frac{2}{3}|\ell_1 - \ell_2|,$$

which is impossible. Therefore,  $\ell_1 = \ell_2$ .  $\square$

**Theorem 4.3 (Algebra of Finite Limits).** Let  $J \subseteq \mathbb{R}$  be an interval, let  $a \in J$ , and let  $f, g : J \setminus \{a\} \rightarrow \mathbb{R}$  be functions. Assume that

$$\lim_{x \rightarrow a} f(x) = \ell_1 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \ell_2,$$

where  $\ell_1, \ell_2 \in \mathbb{R}$ . Let  $\alpha \in \mathbb{R}$ . Then:

1.  $\lim_{x \rightarrow a} (f(x) + g(x)) = \ell_1 + \ell_2$ .
2.  $\lim_{x \rightarrow a} (\alpha f(x)) = \alpha \ell_1$ .
3.  $\lim_{x \rightarrow a} (f(x)g(x)) = \ell_1 \ell_2$ .
4. If  $\ell_1 \neq 0$ , then there exists  $\delta > 0$  such that  $f(x) \neq 0$  whenever  $0 < |x - a| < \delta$ . Moreover,

$$\lim_{x \rightarrow a} \frac{1}{f(x)} = \frac{1}{\ell_1}.$$

**Definition 4.2 (Right-Hand Limit).** Let  $J \subseteq \mathbb{R}$  be an interval and  $a \in J$  be not the right endpoint of  $J$ . Suppose  $f : J \setminus \{a\} \rightarrow \mathbb{R}$  be a function. We say that

$$\lim_{x \rightarrow a^+} f(x) = \ell$$

if for every  $\epsilon > 0$ , there exists  $\delta > 0$  such that

$$x \in J, \quad 0 < x - a < \delta \Rightarrow |f(x) - \ell| < \epsilon.$$

In this case,  $\ell$  is called the **right-hand limit** of  $f$  at  $a$ .

**Definition 4.3 (Left-Hand Limit).** Let  $J \subseteq \mathbb{R}$  be an interval and  $a \in J$  be not the left endpoint of  $J$ . Suppose  $f : J \setminus \{a\} \rightarrow \mathbb{R}$  be a function. We say that

$$\lim_{x \rightarrow a^-} f(x) = \ell$$

if for every  $\epsilon > 0$ , there exists  $\delta > 0$  such that

$$x \in J, \quad 0 < a - x < \delta \Rightarrow |f(x) - \ell| < \epsilon.$$

In this case,  $\ell$  is called the **left-hand limit** of  $f$  at  $a$ .

**Example 4.4.** Let

$$f(x) = \begin{cases} 0, & \text{if } x > 0, \\ 1, & \text{if } x < 0. \end{cases}$$

Then

$$\lim_{x \rightarrow 0^+} f(x) = 0, \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = 1.$$

**Theorem 4.5.** Let  $J \subseteq \mathbb{R}$  be an interval and  $a \in J$  be neither the left endpoint nor the right endpoint of  $J$ . Suppose  $f : J \setminus \{a\} \rightarrow \mathbb{R}$  be a function. Then

$$\lim_{x \rightarrow a} f(x) = \ell$$

if and only if

$$\lim_{x \rightarrow a^-} f(x) = \ell \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = \ell.$$

**Theorem 4.6 (Sequential Characterization of Limits).** Let  $a \in J \subseteq \mathbb{R}$  be an interval and  $f : J \setminus \{a\} \rightarrow \mathbb{R}$  be a function. Then  $\lim_{x \rightarrow a} f(x) = \ell$  if and only if for every sequence  $(x_n)$  in  $J \setminus \{a\}$  satisfying  $x_n \rightarrow a$ , we have  $f(x_n) \rightarrow \ell$ , as  $n \rightarrow \infty$ .

**Example 4.7.** Show that

$$\lim_{x \rightarrow 2} (x^2 + 1) = 5.$$

**Solution.** Let  $(x_n)$  be any sequence such that  $x_n \rightarrow 2$ . Then, by the algebra of convergent sequences,

$$x_n^2 + 1 \rightarrow 2^2 + 1 = 5.$$

Hence, by the Sequential Characterization of Limits,

$$\lim_{x \rightarrow 2} (x^2 + 1) = 5.$$

**Exercise 4.3.** Let

$$f(x) = \begin{cases} 0, & \text{if } x \in \mathbb{Q}, \\ 1, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Show that

$$\lim_{x \rightarrow a} f(x)$$

exists only for  $a = 0$ . *Hint.* Use the density of the rational and irrational numbers in  $\mathbb{R}$ . Alternatively, use the Sequential Characterization of Limits by considering one sequence of rational numbers and one sequence of irrational numbers, both converging to  $a$ .

## 4.2 Infinite Limits

**Definition 4.4 (Infinite Limits).** Let  $a \in J \subseteq \mathbb{R}$  be an interval and  $f : J \setminus \{a\} \rightarrow \mathbb{R}$  be a function.

1. We say that

$$\lim_{x \rightarrow a} f(x) = +\infty$$

if for every  $M > 0$ , there exists  $\delta > 0$  such that

$$x \in J, \quad 0 < |x - a| < \delta \Rightarrow f(x) > M.$$

2. We say that

$$\lim_{x \rightarrow a} f(x) = -\infty$$

if for every  $M < 0$ , there exists  $\delta > 0$  such that

$$x \in J, \quad 0 < |x - a| < \delta \Rightarrow f(x) < M.$$

**Example 4.8.** Show that  $\lim_{x \rightarrow 2} \frac{1}{(x-2)^2} = +\infty$ .

**Solution.** Let  $M > 0$  be given. Choose  $\delta = \frac{1}{\sqrt{M}}$ . If  $0 < |x - 2| < \delta$ , then

$$\frac{1}{(x-2)^2} > M.$$

Hence, by the definition of an infinite limit,

$$\lim_{x \rightarrow 2} \frac{1}{(x-2)^2} = +\infty.$$

**Exercise 4.4.** Solve the following exercises.

1. Use the sequential criterion for limits to show that the following limits do not exist.

(i)  $\lim_{x \rightarrow 0} \cos \frac{\pi}{x}.$

(ii)  $\lim_{x \rightarrow 0} \frac{1}{x} \sin \frac{1}{x}.$

2. Let

$$f(x) = \begin{cases} x, & x \in \mathbb{Q}, \\ -2 - x, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Show that

(i)  $\lim_{x \rightarrow -1} f(x) = 1.$

(ii)  $\lim_{x \rightarrow c} f(x)$

does not exist whenever  $c \neq -1$ .

3. Show that the following limits do not exist.

(i)  $\lim_{x \rightarrow 0} \frac{\sin |x|}{\sin x}.$

(ii)  $\lim_{x \rightarrow 0} \frac{2x + |x|}{x^2 - |x|}.$

4. Evaluate the following limits.

(i)  $\lim_{x \rightarrow 0^+} \sqrt{x - [x]}, \quad \lim_{x \rightarrow 0^-} \sqrt{x - [x]}.$

(iii)  $\lim_{x \rightarrow 0^+} ([x] + [1-x]), \quad \lim_{x \rightarrow 0^-} ([x] + [1-x]).$

(ii)  $\lim_{x \rightarrow 0^+} \frac{1}{e^x + \frac{1}{x}}, \quad \lim_{x \rightarrow 0^-} \frac{1}{e^x + \frac{1}{x}}.$

(iv)  $\lim_{x \rightarrow 0^+} x \left[ \frac{1}{x} \right], \quad \lim_{x \rightarrow 0^-} x \left[ \frac{1}{x} \right].$

5. Evaluate the following limits.

(i)  $\lim_{x \rightarrow \infty} \frac{\sin x}{e^x \cos x}.$

(ii)  $\lim_{x \rightarrow \infty} (x - \sqrt{x^2 - 1}).$

(iii)  $\lim_{x \rightarrow \infty} (\sqrt[3]{x+1} - \sqrt[3]{x}).$

## 5 Continuity

The concept of a limit describes the behavior of a function as the input approaches a point. A natural question is whether the value of the function at that point agrees with the value suggested by its nearby behavior. If this happens at every point in its domain, the function is said to be *continuous*.

Intuitively, a continuous function has no sudden jumps, breaks, or holes in its graph. Small changes in the input produce only small changes in the output. This idea is fundamental in mathematics and its applications, as many physical phenomena, such as the motion of an object, the flow of heat, and population growth, are naturally modeled by continuous functions. The precise definition of continuity is therefore expressed in terms of limits.

**Definition 5.1.** Let  $a \in J \subseteq \mathbb{R}$  and  $f : J \rightarrow \mathbb{R}$  be a function. We say that  $f$  is **continuous at  $a$**  if for every  $\epsilon > 0$ , there exists  $\delta > 0$  such that

$$x \in J \text{ and } |x - a| < \delta \implies |f(x) - f(a)| < \epsilon.$$

We say that  $f$  is **continuous on  $J$**  if it is continuous at every point of  $J$ .

**Example 5.1.** Show that the following functions are continuous.

1.  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = x$ , for all  $x \in \mathbb{R}$ .
2.  $f : [-R, R] \rightarrow \mathbb{R}$ ,  $f(x) = x^2$ , for all  $x \in [-R, R]$ .

**Solution.**

1. Let  $a \in \mathbb{R}$  and let  $\epsilon > 0$  be given. We have

$$|f(x) - f(a)| = |x - a|.$$

Thus, it is natural to choose

$$\delta = \epsilon.$$

Then, whenever  $|x - a| < \delta$ , we obtain

$$|f(x) - f(a)| = |x - a| < \delta = \epsilon.$$

Hence,  $f$  is continuous at  $a$ . Since  $a$  was arbitrary,  $f$  is continuous on  $\mathbb{R}$ .

2. Let  $a \in [-R, R]$  and let  $\epsilon > 0$  be given. Observe that

$$|f(x) - f(a)| = |x^2 - a^2| = |x + a||x - a| \leq (|x| + |a|)|x - a| \leq 2R|x - a|.$$

Choose

$$\delta = \frac{\epsilon}{2R}.$$

Then, whenever  $|x - a| < \delta$ , we have

$$|f(x) - f(a)| \leq 2R|x - a| < 2R\delta = \epsilon.$$

Hence,  $f$  is continuous on  $[-R, R]$ .

**Exercise 5.1.** Show that the function  $f : \mathbb{R} \rightarrow \mathbb{R}$ , defined by  $f(x) = x^2$ , is continuous on  $\mathbb{R}$  using the  $\epsilon$ - $\delta$  definition. Hint. Observe that  $|x^2 - a^2| = |x - a||x + a|$ . To estimate  $|x + a|$ , first assume  $\delta \leq 1$ . Then, whenever  $|x - a| < \delta$ , we have

$$|x| \leq |x - a| + |a| < 1 + |a|,$$

and hence

$$|x + a| \leq |x| + |a| < 1 + 2|a|.$$

Now, choose  $\delta = \min \left\{ 1, \frac{\epsilon}{1+2|a|} \right\}$ .

## 5.1 Sequential Criterion for Continuity

**Definition 5.2.** Let  $a \in J \subseteq \mathbb{R}$  and  $f : J \rightarrow \mathbb{R}$  be a function. We say that  $f$  is **continuous at**  $a$  if for every sequence  $(x_n)$  in  $J$  with  $x_n \rightarrow a$ , we have  $f(x_n) \rightarrow f(a)$ . We say that  $f$  is **continuous on**  $J$  if it is continuous at every point of  $J$ .

**Example 5.2.** 1. Show that every constant function is continuous.

2. Show that the identity function  $f : J \rightarrow \mathbb{R}$ , defined by  $f(x) = x$  for all  $x \in J$ , is continuous.

3. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Then  $f$  is not continuous at 0.

**Solution.**

1. Let  $J \subseteq \mathbb{R}$  and  $f : J \rightarrow \mathbb{R}$  be the constant function defined by  $f(x) = c$  for all  $x \in J$ .

Fix any  $a \in J$ , and  $(x_n)$  be a sequence in  $J$  such that  $x_n \rightarrow a$ . Since  $f(x_n) = c$  for every  $n$ , the sequence  $(f(x_n))$  is constant. Hence,

$$f(x_n) \rightarrow c = f(a).$$

Therefore,  $f$  is continuous at  $a$ . As  $a$  was arbitrary, we conclude that  $f$  is continuous on  $J$ .

2. Fix any  $a \in J$ , and let  $(x_n)$  be a sequence in  $J$  such that  $x_n \rightarrow a$ . Since  $f(x_n) = x_n$  for every  $n$ , we have

$$f(x_n) = x_n \rightarrow a = f(a).$$

Hence,  $f$  is continuous at  $a$ . Since  $a$  was arbitrary,  $f$  is continuous on  $J$ .

More generally, for every  $n \in \mathbb{N}$ , the function  $f : \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = x^n$  is continuous.

3. Consider the sequence  $x_n = \frac{1}{n}$ . Then  $x_n \in \mathbb{Q}$  and  $x_n \rightarrow 0$ . Hence

$$f(x_n) = 1 \rightarrow 1 = f(0).$$

Now let

$$y_n = \frac{\sqrt{2}}{n}.$$

Since  $\sqrt{2}$  is irrational,  $y_n$  is irrational for every  $n$ . Also,  $y_n \rightarrow 0$ . Therefore,

$$f(y_n) = 0 \longrightarrow 0 \neq 1 = f(0).$$

Thus, by the sequential criterion for continuity,  $f$  is not continuous at 0. Using the density of the rational and irrational numbers in  $\mathbb{R}$ , together with the Sandwich Theorem, one can similarly show that  $f$  is not continuous at any point of  $\mathbb{R}$ .

**Example 5.3.** Consider the function

$$f(x) = \begin{cases} 1, & \text{if } x > 0, \\ 0, & \text{if } x \leq 0. \end{cases}$$

We claim that  $f$  is continuous at every  $a \neq 0$  but is not continuous at 0. Indeed, let  $a \neq 0$  and let  $(x_n)$  be a sequence converging to  $a$ . Since  $a$  is either positive or negative, there exists  $N \in \mathbb{N}$  such that  $x_n$  has the same sign as  $a$  for all  $n \geq N$ . Consequently,  $f(x_n) = f(a)$  for all sufficiently large  $n$ , and hence

$$f(x_n) \longrightarrow f(a).$$

Thus  $f$  is continuous at every  $a \neq 0$ . Now let  $a = 0$ . Consider the sequences

$$x_n = \frac{1}{n}, \quad y_n = -\frac{1}{n}.$$

Both converge to 0, but

$$f(x_n) = 1 \longrightarrow 1, \quad f(y_n) = 0 \longrightarrow 0.$$

Since the limits are different,  $f$  is not continuous at 0.

**Example 5.4.** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be defined by

$$f(x) = \begin{cases} c, & \text{if } x < 0, \\ \alpha, & \text{if } x = 0, \\ c, & \text{if } x > 0. \end{cases}$$

Determine the value of  $\alpha$  for which  $f$  is continuous at 0.

*Proof.* Suppose that  $f$  is continuous at 0. Consider the sequence

$$x_n = -\frac{1}{n}.$$

Then  $x_n \rightarrow 0$  and

$$f(x_n) = c \longrightarrow f(0) = \alpha.$$

Hence,

$$\alpha = c.$$

Conversely, assume that  $\alpha = c$ . Let  $(x_n)$  be any sequence with  $x_n \rightarrow 0$ . If  $x_n < 0$ , then

$$f(x_n) = c = f(0).$$

If  $x_n > 0$ , then

$$f(x_n) = c = f(0).$$

In either case,  $f(x_n) \rightarrow c = f(0)$ . Hence,  $f$  is continuous at 0. Thus,  $f$  is continuous at 0 if and only if  $\alpha = c$ .  $\square$

**Theorem 5.5.** The sequential definition of continuity and the  $\epsilon$ - $\delta$  definition are equivalent.

*Proof.* Left as an exercise. □

**Theorem 5.6 (Algebra of Continuous Functions).** *Let  $J \subseteq \mathbb{R}$  and  $f, g : J \rightarrow \mathbb{R}$  be continuous at  $a \in J$ . Suppose  $\alpha \in \mathbb{R}$ . Then:*

1.  $f + g$  is continuous at  $a$ .
2.  $\alpha f$  is continuous at  $a$ .
3.  $fg$  is continuous at  $a$ .
4. If  $g(a) \neq 0$ , then there exists  $\delta > 0$  such that  $g(x) \neq 0$  for all  $x \in (a - \delta, a + \delta) \cap J$ . Consequently,  $\frac{1}{g}$  is continuous at  $a$ , and hence so is  $\frac{f}{g}$ .
5.  $|f|$  is continuous at  $a$ .
6. The functions  $\max\{f, g\}$  and  $\min\{f, g\}$  are continuous at  $a$ .
7. If  $K \subseteq \mathbb{R}$ ,  $h : K \rightarrow \mathbb{R}$  is continuous at  $f(a)$ , and  $f(J) \subseteq K$ , then the composition  $h \circ f$  is continuous at  $a$ .

**Remark 5.7.** *If  $|f|$  is continuous on  $J$ , then  $f$  need not be continuous on  $J$ . For example, let  $J = \mathbb{R}$  and define*

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ -1, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

*Then  $|f| : \mathbb{R} \rightarrow \mathbb{R}$  is given by  $|f|(x) = 1$ ,  $x \in \mathbb{R}$ . Hence  $|f|$  is continuous on  $\mathbb{R}$ , whereas  $f$  is not continuous on  $\mathbb{R}$ .*

## 6 Discontinuity

We have seen that a function defined on a subset  $J$  of  $\mathbb{R}$  may be continuous at every point of  $J$ , continuous only at some points, or discontinuous at one or more points of  $J$ . The behavior of a function near a point determines the nature of its discontinuity.

Different discontinuities arise in different ways. To understand and classify them, we now study the various types of discontinuities of a function.

### 6.1 Types of Discontinuities

1.  $\lim_{x \rightarrow a^-} f(x)$  does not exist.
2.  $\lim_{x \rightarrow a^+} f(x)$  does not exist.
3.  $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$ : **Jump discontinuity**
4.  $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) \neq f(a)$ : **Removable discontinuity**

**Example 6.1.** 1. Consider the function

$$f(x) = \sin\left(\frac{1}{x}\right), \quad x \neq 0.$$

Then  $\lim_{x \rightarrow 0^-} f(x)$  does not exist.

2. Consider the function

$$f(x) = \frac{1}{x}, \quad x \neq 0.$$

Then  $\lim_{x \rightarrow 0^-} f(x)$  does not exist (it is  $-\infty$ ).

3. Consider the function

$$f(x) = \begin{cases} 1, & \text{if } x > 0, \\ 0, & \text{if } x < 0. \end{cases}$$

Then

$$\lim_{x \rightarrow 0^-} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^+} f(x) = 1.$$

Since the left-hand and right-hand limits are different,  $f$  is discontinuous at 0.

4. Consider the function

$$f(x) = \begin{cases} 1, & \text{if } x \neq 0, \\ 2026, & \text{if } x = 0. \end{cases}$$

Then

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = 1,$$

but  $f(0) = 2026$ . Hence  $f$  is discontinuous at 0.

5. Consider the function

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Since every interval contains both rational and irrational numbers, neither

$$\lim_{x \rightarrow 0^-} f(x) \quad \text{nor} \quad \lim_{x \rightarrow 0^+} f(x)$$

exists. Hence  $f$  is discontinuous at 0. Indeed, the function is discontinuous at every point of  $\mathbb{R}$ .

6. Define

$$f(x) = \begin{cases} x, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Then  $f$  is continuous only at  $x = 0$  and is discontinuous at every other point of  $\mathbb{R}$ .

## 6.2 Properties of Continuous Functions

Suppose  $f : [a, b] \rightarrow \mathbb{R}$  is a continuous function such that  $f(a)$  and  $f(b)$  have opposite signs. Draw the graphs of a few such functions, considering both the cases  $f(a) > 0 > f(b)$  and  $f(a) < 0 < f(b)$ .

What do you observe? Does the graph always intersect the  $x$ -axis? If so, can you explain why this must happen? In other words, must there exist a point  $c \in (a, b)$  such that  $f(c) = 0$ ?

The answer is affirmative, and it is one of the most important consequences of continuity. This result is known as the Intermediate Value Theorem.

**Theorem 6.2** (Cantor's Nested Interval Theorem). *Let  $\{[a_n, b_n]\}$  be a sequence of closed intervals such that*

1.  $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$  for all  $n \in \mathbb{N}$ ,

$$2. \lim_{n \rightarrow \infty} (b_n - a_n) = 0.$$

Then  $\bigcap_{n=1}^{\infty} [a_n, b_n]$  contains exactly one point.

**Theorem 6.3** (Bolzano). *Let  $[a, b]$  be a closed and bounded interval and  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$ . If  $f(a)$  and  $f(b)$  are of opposite signs then there exists at least a point  $c$  in the open interval  $(a, b)$  such that  $f(c) = 0$ .*

*Proof.* Let  $I_1 = [a, b] = [a_1, b_1]$ , say. Without loss of generality, let us assume that  $f(a_1) < 0$  and  $f(b_1) > 0$ .

Let  $c_1 = \frac{a_1 + b_1}{2}$ . Then  $f(c_1)$  is either 0 or  $\neq 0$ . If  $f(c_1) = 0$  the theorem is proved. If  $f(c_1) \neq 0$  then either  $f(c_1) < 0$ , or  $f(c_1) > 0$ .

If  $f(c_1) < 0$  we consider the closed interval  $[c_1, b_1]$  and call it  $I_2 = [a_2, b_2]$ . If  $f(c_1) > 0$  we consider the closed interval  $[a_1, c_1]$  and call it  $I_2 = [a_2, b_2]$ .

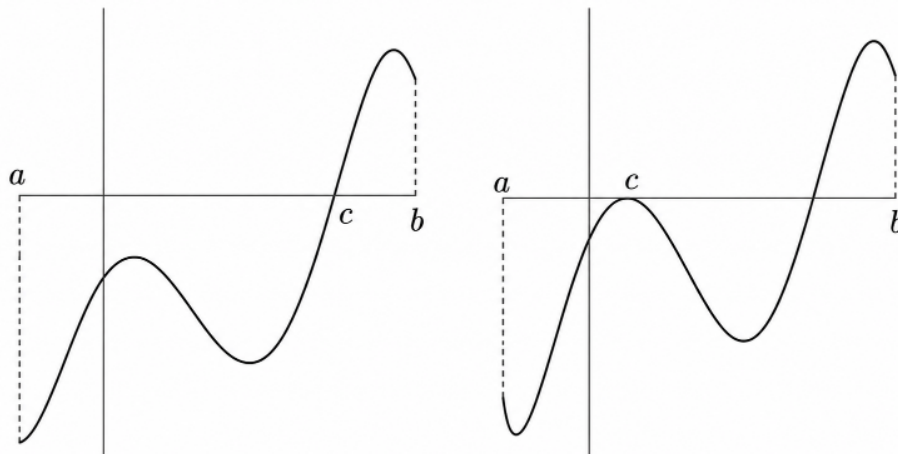
Thus if  $f(c_1) \neq 0$ , the closed interval  $I_2 = [a_2, b_2]$  is such that (i)  $f$  is continuous on  $I_2$  and  $f(a_2) < 0$ ,  $f(b_2) > 0$ ; (ii)  $I_2 \subset I_1$ ; (iii)  $|I_2| = \frac{1}{2}(b - a)$ .

Let  $c_2 = \frac{a_2 + b_2}{2}$ . Then  $f(c_2)$  is either 0 or  $\neq 0$ .

If  $f(c_2) = 0$  the theorem is proved. If  $f(c_2) \neq 0$ , then either  $f(c_2) < 0$ , or  $f(c_2) > 0$ .

If  $f(c_2) < 0$  we consider the closed interval  $[c_2, b_2]$  and call it  $I_3 = [a_3, b_3]$ . If  $f(c_2) > 0$  we consider the closed interval  $[a_2, c_2]$  and call it  $I_3 = [a_3, b_3]$ .

Thus if  $f(c_2) \neq 0$  the closed interval  $I_3 = [a_3, b_3]$  is such that (i)  $f$  is continuous on  $I_3$  and  $f(a_3) < 0$ ,  $f(b_3) > 0$ ; (ii)  $I_3 \subset I_2 \subset I_1$ ; (iii)  $|I_3| = \frac{b-a}{2^2}$ .



Let  $c_3 = \frac{a_3 + b_3}{2}$ . Then  $f(c_3)$  is either 0 or  $\neq 0$ .

Continuing in this manner, either we obtain a point  $c_k$  in  $(a, b)$  such that  $f(c_k) = 0$ , in which case the theorem is proved, or we obtain a sequence of closed and bounded intervals  $\{I_n\}$  such that (i)  $f$  is continuous on  $I_n$  and  $f(a_n) < 0$ ,  $f(b_n) > 0$  for all  $n \in \mathbb{N}$ ; (ii)  $I_{n+1} \subset I_n$  for all  $n \in \mathbb{N}$ ; (iii)  $|I_n| = \frac{b-a}{2^n}$  and therefore  $\lim |I_n| = 0$ .

Thus  $\{I_n\}$  is a sequence of nested closed and bounded intervals with  $\lim |I_n| = 0$ .

By nested intervals theorem there exists one and only one point  $\alpha$  such that (i)  $\alpha \in [a_n, b_n]$  for all  $n \in \mathbb{N}$ , and (ii)  $\lim a_n = \alpha = \lim b_n$ .

From (i) it follows that  $\alpha \in [a, b]$ . Since  $f$  is continuous on  $[a, b]$ ,  $f$  is continuous at  $\alpha$ .

From (ii) it follows that  $\{a_n\}$  is a sequence of points in  $[a, b]$  converging to  $\alpha$ . Since  $f$  is continuous at  $\alpha$ ,  $\lim f(a_n) = f(\alpha)$ .

But  $f(a_n) < 0$  for all  $n \in \mathbb{N}$  and this implies  $\lim f(a_n) \leq 0$ .

That is,

$$f(\alpha) \leq 0 \quad (\text{A})$$

Also from (ii) it follows that  $\{b_n\}$  is a sequence of points in  $[a, b]$  converging to  $\alpha$ . Since  $f$  is continuous at  $\alpha$ ,  $\lim f(b_n) = f(\alpha)$ .

But  $f(b_n) > 0$  for all  $n \in \mathbb{N}$  and this implies  $\lim f(b_n) \geq 0$ .

That is,

$$f(\alpha) \geq 0 \quad (\text{B})$$

From (A) and (B) it follows that  $f(\alpha) = 0$ .

Since  $\alpha \in [a, b]$  and  $f(a) < 0$ ,  $f(b) > 0$  it follows that  $\alpha \in (a, b)$ .

Thus  $\alpha = c$  and the theorem is proved.  $\square$

**Theorem 6.4 (Intermediate Value Theorem).** *Let  $g : [a, b] \rightarrow \mathbb{R}$  be a continuous function, and let  $\lambda$  be a real number between  $g(a)$  and  $g(b)$ . Then there exists a point  $c \in (a, b)$  such that*

$$g(c) = \lambda.$$

*Proof.* Without loss of generality, assume that  $g(a) < \lambda < g(b)$ . Define

$$f(x) = g(x) - \lambda.$$

Since  $g$  is continuous on  $[a, b]$ , so is  $f$ . Moreover,

$$f(a) = g(a) - \lambda < 0 \quad \text{and} \quad f(b) = g(b) - \lambda > 0.$$

Hence, by Theorem 6.3, there exists  $c \in (a, b)$  such that  $f(c) = 0$ . Therefore,  $g(c) = \lambda$ .  $\square$

**Remark 6.5.** *Let  $I = [a, b]$  be a closed and bounded interval and  $f : [a, b] \rightarrow \mathbb{R}$  be such that  $f(a) \neq f(b)$  and  $f$  attains every value between  $f(a)$  and  $f(b)$  at least once in  $(a, b)$ . Still  $f$  may not be continuous on  $[a, b]$ .*

*For example, let  $f : [0, 2] \rightarrow \mathbb{R}$  be defined by  $f(0) = 0$ ,  $f(2) = 2$  and*

$$f(x) = \begin{cases} x, & 0 < x \leq 1, \\ 3 - x, & 1 < x < 2. \end{cases}$$

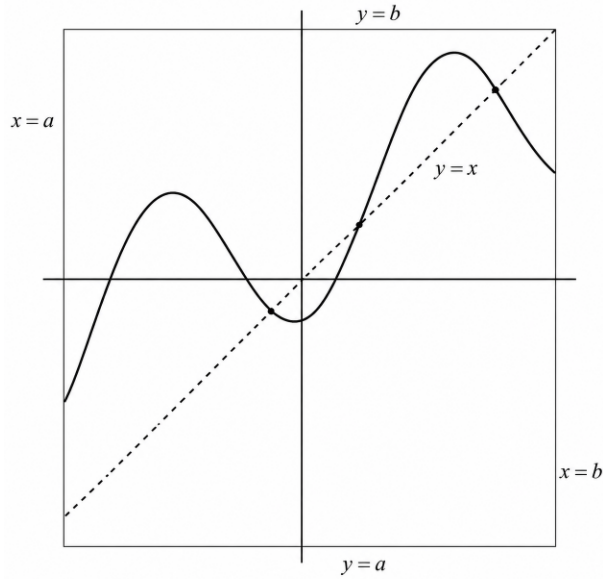
*$f$  assumes every value between 0 and 2 on  $[0, 2]$ . But  $f$  is not continuous on  $[0, 2]$  since  $f$  is not continuous at 1 and 2.*

**Theorem 6.6 (Fixed Point Theorem).** *Let  $f : [a, b] \rightarrow [a, b]$  be a continuous function. Then there exists  $c \in [a, b]$  such that  $f(c) = c$ . Such a point  $c$  is called a **fixed point** of  $f$ .*

*Proof.* Define  $g(x) = f(x) - x$ . Since  $g$  is continuous on  $[a, b]$ , we have  $g(a) = f(a) - a \geq 0$  and  $g(b) = f(b) - b \leq 0$ . If either  $g(a) = 0$  or  $g(b) = 0$ , then we are done. Otherwise,  $g(a) > 0 > g(b)$ . Hence, by the Intermediate Value Theorem, there exists  $c \in (a, b)$  such that  $g(c) = 0$ . Therefore,  $f(c) = c$ .  $\square$

**Example 6.7.** 1. *Show that the equation  $xe^x = 1$  has a solution in the interval  $(0, 1)$ .*

2. *Let  $f : [0, 2\pi] \rightarrow \mathbb{R}$  be a continuous function satisfying  $f(0) = f(2\pi)$ . Show that there exists  $x \in [0, \pi]$  such that  $f(x) = f(x + \pi)$ .*



3. Show that every real polynomial of odd degree defines an onto function from  $\mathbb{R}$  to  $\mathbb{R}$ .

**Solution.**

1. Define  $f : [0, 1] \rightarrow \mathbb{R}$  by  $f(x) = xe^x - 1$ . Since  $f$  is continuous on  $[0, 1]$ , we have  $f(0) = -1 < 0$  and  $f(1) = e - 1 > 0$ . Hence, by the Intermediate Value Theorem, there exists  $c \in (0, 1)$  such that  $f(c) = 0$ . Therefore,  $ce^c = 1$ .

2. Define  $g : [0, \pi] \rightarrow \mathbb{R}$  by  $g(x) = f(x) - f(x + \pi)$ . Since  $f$  is continuous, so is  $g$ . Moreover,

$$g(0) = f(0) - f(\pi) \quad \text{and} \quad g(\pi) = f(\pi) - f(2\pi) = f(\pi) - f(0) = -g(0).$$

If  $g(0) = 0$ , then  $f(0) = f(\pi)$  and we are done. Otherwise,  $g(0)$  and  $g(\pi)$  have opposite signs. Hence, by the Intermediate Value Theorem, there exists  $c \in (0, \pi)$  such that  $g(c) = 0$ . Therefore,  $f(c) = f(c + \pi)$ .

3. Let  $p : \mathbb{R} \rightarrow \mathbb{R}$  be a polynomial of odd degree, and let  $y \in \mathbb{R}$  be arbitrary. We wish to show that there exists  $x \in \mathbb{R}$  such that  $p(x) = y$ . Define the function  $f : \mathbb{R} \rightarrow \mathbb{R}$  by  $f(x) = p(x) - y$ . Since  $f$  is continuous, it suffices to show that  $f$  changes sign. As  $p$  has odd degree, we have  $\lim_{x \rightarrow \infty} p(x) = \infty$  and  $\lim_{x \rightarrow -\infty} p(x) = -\infty$ , or vice versa, depending on the sign of the leading coefficient. Hence, there exist  $a, b \in \mathbb{R}$  such that  $f(a) < 0 < f(b)$ . By the Intermediate Value Theorem, there exists  $c \in (a, b)$  such that  $f(c) = 0$ . Therefore,  $p(c) = y$ . Since  $y$  was arbitrary,  $p : \mathbb{R} \rightarrow \mathbb{R}$  is onto.

**Definition 6.1.** Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . We say that  $f$  is **bounded on**  $A$  if there exists  $M > 0$  such that  $|f(x)| \leq M$  for every  $x \in A$ .

**Example 6.8.** 1. The function  $f : \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = x$  is not bounded. Indeed, for every  $M > 0$ , choosing  $x = M + 1$  gives  $f(x) = M + 1 > M$ . Hence,  $f$  is unbounded.

2. The function  $f : \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = \sin x$  is bounded. In fact,  $|f(x)| \leq 1$  for every  $x \in \mathbb{R}$ .

**Theorem 6.9 (Weierstrass Theorem).** Let  $f : [a, b] \rightarrow \mathbb{R}$  be a continuous function. Then  $f$  is bounded.

*Sketch of the proof.* The proof is based on the same least upper bound argument used in the proof of the Intermediate Value Theorem.

Consider the set of all points  $x \in [a, b]$  such that  $f$  is bounded on the interval  $[a, x]$ . Since  $f$  is continuous, it is bounded in a neighborhood of  $a$ . Thus, this set is nonempty and bounded above by  $b$ . Let  $c$  be its least upper bound.

If  $c < b$ , then continuity of  $f$  at  $c$  implies that  $f$  is bounded in a neighborhood of  $c$ . Consequently,  $f$  is bounded on a larger interval  $[a, c_1]$  for some  $c_1 > c$ , contradicting the definition of  $c$ . Hence,  $c = b$ , and therefore  $f$  is bounded on  $[a, b]$ .  $\square$

**Theorem 6.10 (Extreme Value Theorem).** *Let  $f : [a, b] \rightarrow \mathbb{R}$  be a continuous function. Then there exist points  $x_{\min}, x_{\max} \in [a, b]$  such that*

$$f(x_{\min}) \leq f(x) \leq f(x_{\max})$$

*for every  $x \in [a, b]$ .*

*Proof.* By the Weierstrass Theorem, the set  $A = \{f(x) : x \in [a, b]\}$  is bounded. Let  $M = \sup A$ .

For each  $n \in \mathbb{N}$ , choose  $x_n \in [a, b]$  such that  $M - \frac{1}{n} < f(x_n) \leq M$ . Then  $f(x_n) \rightarrow M$ . Since  $(x_n)$  is a bounded sequence, the Bolzano-Weierstrass Theorem yields a subsequence  $(x_{n_k})$  converging to some  $x \in [a, b]$ . By continuity,  $f(x_{n_k}) \rightarrow f(x)$ . Hence,  $f(x) = M$ .

Similarly, if  $m = \inf A$ , there exists  $y \in [a, b]$  such that  $f(y) = m$ . Therefore,  $f$  attains both its maximum and minimum values on  $[a, b]$ .  $\square$

**Example 6.11.** 1. *Show that there is no continuous function  $f : [0, 1] \rightarrow (0, \infty)$  which is onto.*

2. *Show that there is no continuous function  $f : [a, b] \rightarrow (0, 1)$  which is onto.*

**Solution.**

1. *Suppose such a function exists. Since  $f$  is continuous on the closed interval  $[0, 1]$ , the Weierstrass Theorem implies that  $f$  is bounded. Hence there exists  $M > 0$  such that  $f(x) \leq M$  for every  $x \in [0, 1]$ . Therefore, no number greater than  $M$  belongs to the range of  $f$ , which contradicts the assumption that  $f$  is onto.*

2. *Suppose such a function exists. By the Extreme Value Theorem, there exist  $x_{\min}, x_{\max} \in [a, b]$  such that*

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}), \quad \forall x \in [a, b].$$

*Since the range of  $f$  is  $(0, 1)$ , we have  $0 < f(x_{\min})$  and  $f(x_{\max}) < 1$ . Hence, neither 0 nor 1 belongs to the range of  $f$ , contradicting the assumption that  $f$  is onto.*